

Descriptive Aspects of Structures with Countable Automorphism Groups

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1 Question

We consider infinite countable first-order structures, their automorphism groups and the descriptive aspects of their representations. A representation $\mathcal{A}$ of such a structure is an instance of its isomorphism type in which the universe or domain of the structure is the set of natural numbers $\mathbb{N}$ and the constants, functions and relations of the structure are elements of $\mathbb{N}$, multi-variable functions from $\mathbb{N}$ to $\mathbb{N}$ and multi-place relations on $\mathbb{N}$, respectively. Similarly, an automorphism of $\mathcal{A}$ or an isomorphism from $\mathcal{A}$ to $\mathcal{B}$ is a bijective function from $\mathbb{N}$ to $\mathbb{N}$ that preserves the evaluation of all constants, functions and relations. We let $AUT(\mathcal{A})$ denote the automorphism group of $\mathcal{A}$.

Definition 1.1. • $\mathcal{A}$ is *computably AUT -countable* if every automorphism of $\mathcal{A}$ is computable relative to $\mathcal{A}$.

- $\mathcal{A}$ is *computably AUT -countable on a cone* if there is a $C \subset \mathbb{N}$ such that whenever $\mathcal{B}$ is isomorphic to $\mathcal{A}$, every automorphism of $\mathcal{B}$ is computable relative to $\mathcal{B} \oplus C$.

Definition 1.2. A function $f : \mathcal{A} \rightarrow \mathcal{A}$ is Σ_1^{in} -*definable* in $\mathcal{A}$ relative to parameters $\bar{a} = (a_1, \dots, a_n)$ from $\mathcal{A}$ if there is a family of existential formulas $(\varphi_i)_{i \in \mathbb{N}}$ such that for all x and y in $\mathcal{A}$, $f(x) = y$ if and only if there is an $i \in \mathbb{N}$ such that $\mathcal{A} \models \varphi_i(\bar{a}, x, y)$.

Question 1.3. Is there a countable computably-represented structure $\mathcal{A}$ such that $\mathcal{A}$ is computably AUT -countable on a cone and such that for any

finite set of parameters $\bar{a}$ in $\mathcal{A}$ there is an automorphism π of $\mathcal{A}$ such that π is not Σ_1^{in} -definable in $\mathcal{A}$ relative to the parameters $\bar{a} \cup \pi(\bar{a})$, where $\pi(\bar{a})$ is the image of $\bar{a}$ under π .

2 Background

We note some examples.

- Example 2.1.**
- Any rigid structure, having no nontrivial automorphism, is computably *AUT*-countable, since the identity map is computable no matter how complicated the structure may be.
 - The first order structure represented by the integers with the successor function is computably *AUT*-countable, since every automorphism is a finite iteration of the successor or of its inverse.

We recall a fundamental model theoretic result.

Theorem 2.2 (see [2, Cor. 4.2.5]). *Let $\mathcal{A}$ be a countable structure and suppose that $AUT(\mathcal{A})$ is countable. There are finitely many parameters $\bar{a} \in \mathcal{A}$ such that every automorphism of $\mathcal{A}$ is determined by its action on $\bar{a}$.*

Theorem 2.3 (following [1]). *Let $\mathcal{A}$ be a countable structure. $\mathcal{A}$ is computably *AUT*-countable on a cone if and only if for each $\pi \in AUT(\mathcal{A})$, there are finitely many parameters $\bar{a} \in \mathcal{A}$ such that π is Σ_1^{in} -definable in $\mathcal{A}$ relative to $\bar{a}$.*

[1] showed that for a relation R on $\mathcal{A}$, R is computably enumerable on a cone if and only if there are finitely many parameters $\bar{a} \in \mathcal{A}$ and a family of existential formulas $(\varphi_i)_{i \in \mathbb{N}}$ such that all $\bar{x} \subset \mathcal{A}$, $\bar{x} \in R$ if and only if there is an i such that $\mathcal{A} \models \varphi_i(\bar{a}, \bar{x})$. We obtain Theorem 2.3 by applying this result to the graphs of the automorphisms of $\mathcal{A}$.

When its hypotheses apply, Theorem 2.3 gives a specific syntactic form such that if π is an automorphism of $\mathcal{A}$, then there are finitely many parameters within $\mathcal{A}$ such that the action of π is definable from those parameters. In the result as stated, the parameters used to define π depend on which π is being defined. This does not address the full content of Theorem 2.2, which asserts that there are finitely many parameters such that every automorphism is determined by its action on those parameters. In Question 1.3, we

ask whether there is a counter-example to what would have been a natural effective-ization of Theorem 2.2, that there would be one finite set of parameters from which every automorphism could be effectively defined.

3 Solution

Theorem 3.1. *There is a computably-represented structure $\mathcal{A}$ such that $\mathcal{A}$ is computably AUT-countable on a cone and such that for any finite set of parameters $\bar{a}$ there is an automorphism π of $\mathcal{A}$ such that π is not Σ_1^{in} -definable in $\mathcal{A}$ relative to $\bar{a} \cup \pi(\bar{a})$.*

Typically, counter-examples to computability-theoretic statements are produced by diagonal constructions. This is not the case here. Rather, the structure exhibited in Theorem 3.1 comes from properties of elementary arithmetic, specifically those of $(\mathbb{Q}, +)$, the additive group on the rational numbers.

As is well-known, π is an automorphism of $(\mathbb{Q}, +)$ if and only if there is a non-zero rational s such that for all x , $\pi(x) = sx$. Equivalently, the automorphism group of $(\mathbb{Q}, +)$ is naturally isomorphic to the multiplicative group on the non-zero rational numbers, conventionally denoted as $(\mathbb{Q}^*, \cdot)$. Here is a quick reminder of the proof. To verify the non-trivial implication, suppose that π is an automorphism of $(\mathbb{Q}, +)$ and let a/b be the reduced-form fraction that represents $\pi(1)$. Consider a non-zero rational x represented by c/d . Then

$$\underbrace{x + \cdots + x}_{d\text{-times}} = \underbrace{1 + \cdots + 1}_{c\text{-times}}$$

and so

$$\underbrace{\pi(x) + \cdots + \pi(x)}_{d\text{-times}} = \underbrace{a/b + \cdots + a/b}_{c\text{-times}}.$$

It follows that $\pi(x) = \frac{ca/b}{d} = a/b \cdot c/d = a/b \cdot x$. For the case when $x = 0$, $\pi(0) = 0 = a/b \cdot 0$, as required to show for all x , $\pi : x \mapsto a/b \cdot x$.

Before giving the proof of Theorem 3.1, we note that $(\mathbb{Q}, +)$ is not an example of the desired sort. If $\pi : x \mapsto a/b \cdot x$, then $\pi(x) = y$ exactly when

$$(\mathbb{Q}, +) \models \underbrace{y + \cdots + y}_{b\text{-times}} = \underbrace{x + \cdots + x}_{a\text{-times}},$$

which shows that the graph of π is definable in $(\mathbb{Q}, +)$ without parameters by a formula without quantifiers. We produce an example $\mathcal{A}$ to verify Theorem 3.1, by rearranging the arithmetic of $(\mathbb{Q}, +)$.

Proof of Theorem 3.1. The language of $\mathcal{A}$ consists of two unary predicates N and Q , a binary relation symbol $<$ and a ternary relation symbol R . The structure $\mathcal{A}$ consists of

1. A copy of the rational numbers, which constitutes the elements of $\mathcal{A}$ which satisfy Q in $\mathcal{A}$.
2. A copy of the natural numbers, disjoint from Q , which constitutes the elements of $\mathcal{A}$ which satisfy N in $\mathcal{A}$.
3. In $\mathcal{A}$, $n < m$ if and only if n and m are in N and n is less than m in the natural order.
4. Fix a computable counting $S_0, S_1, \dots$ of the finite subsets of the non-zero rational numbers so that for each such finite set S there are infinitely many n such that $S_n = S$. In $\mathcal{A}$, $R(n, x, y)$ holds if and only if $n \in N$ and there is an $s \in S_n$ such that $y = sx$, equivalently $y/x \in S_n$, for x nonzero.

To simplify notation, we identify N , Q , $<$ and R with their interpretations in $\mathcal{A}$.

Representing elements of $AUT(\mathcal{A})$. We claim that, just as for $(\mathbb{Q}, +)$, the automorphisms of $\mathcal{A}$ are represented by scalar multiplication by non-zero rational constants. By inspection, if s is a nonzero rational number, then the map on Q given by $x \mapsto sx$ induces an automorphism of $\mathcal{A}$ (which is the identity on N). For the converse, suppose that π is an automorphism of $\mathcal{A}$. We claim that there is a non-zero rational number s such that π is induced by the map $x \mapsto sx$ on Q .

To prove the claim, begin by noting that since every element of N is definable, π fixes the elements of N . Consequently, π fixes the sets S_n which define the relation R .

Next, we show that addition on Q can be evaluated using the first order structure of $\mathcal{A}$. Suppose that a and b are elements of Q . If $a = 0$, an $\mathcal{A}$ -definable condition, then $a + b = b$. Otherwise, begin to evaluate $a + b$ by letting n be least such that S_n is a singleton and $R(n, a, b)$. In other words,

$S_n = \{s\}$ and $b = sa$. Next, let m be least such that $S_m = \{1 + s\}$. Finally, evaluate $a + b$ by noting that $a + b = c$ if and only if $R(m, a, c)$. Thus, π fixes the additive structure on Q . As noted earlier, the automorphisms of $(\mathbb{Q}, +)$ are exactly the functions $x \mapsto sx$, for s a non-zero rational number, and the claim follows.

Computable *AUT*-countability on cone. Suppose that $\mathcal{A}^*$ is computable from X and $f : \mathcal{A}^* \xrightarrow{\sim} \mathcal{A}$. Then, an isomorphism π^* of $\mathcal{A}^*$ induces an isomorphism π of $\mathcal{A}$. As noted π is induced by scalar multiplication by a rational s , there is an $n \in N$ such that for all x and y , $\pi(x) = y$ if and only if $\mathcal{A} \models R(n, x, y)$. Consequently, for x^* and y^* in $\mathcal{A}^*$, $\pi^*(x^*) = y^*$ if and only if $\pi(f(x^*)) = f(y^*)$ if and only if $\mathcal{A} \models R(n, f(x^*), f(y^*))$ if and only if $\mathcal{A}^* \models R(n, x^*, y^*)$. So, π^* is defined in $\mathcal{A}^*$ by a quantifier-free formula relative to the parameter n and thereby computable from X . It follows that $\mathcal{A}$ is computably *AUT*-countable on a cone, indeed on the cone above $\emptyset$.

Contradicting existential definability of automorphisms. The argument which follows is a generalization of the one that shows that $(\mathbb{Q}^*, \cdot)$ is not finitely generated. To expedite the discussion, we say that a prime number p is free over a finite set F of rational numbers if and only if for every element of x of F , if a/b is the reduced-form fraction that represents x then p is not a factor of either a or b . We use the notation pF to denote the set $\{px : x \in F\}$.

Suppose $\bar{c} = \{c_1, \dots, c_k\}$ is a finite subset of Q and $\bar{n} = \{1, \dots, n\}$ is a finite initial segment of N . We may assume that 0 is an element of $\bar{c}$. We propose to show that there is an automorphism π of $\mathcal{A}$, given by multiplication by a non-zero rational number, such that π is not definable by a countable family of existential formulas in the parameters $\bar{n}$, $\bar{c}$ and the images of $\bar{c}$ under π . We will show this in a strong form. Namely, we will show that if p and q are sufficiently free distinct prime numbers and φ is an existential formula such that $\varphi(\bar{n}, \bar{c}, p\bar{c}, q, pq)$ holds in $\mathcal{A}$, then there is a number y which is not equal to pq such that $\varphi(\bar{n}, \bar{c}, p\bar{c}, q, y)$ also holds in $\mathcal{A}$. The number y will be obtained by multiplying pq by a prime which is free over everything that came before. Note that this is sufficient to prove Theorem 3.1.

With these comments in mind, let φ be an existential formula in the language of $\mathcal{A}$. Let p be a prime number which is free over $\bar{c}$ and over all the $s_j \in S_m$, for $m \in \bar{n}$. We examine the automorphism of $\mathcal{A}$ given by

multiplication within Q by p . Let q be a prime which is free over $p, \bar{c}$, and all the $s_j \in S_m$, for $m \in \bar{n}$. Suppose that $\mathcal{A} \models \varphi(\bar{n}, \bar{c}, p\bar{c}, q, pq)$, ostensibly to witness in $\mathcal{A}$ that multiplication by p maps q to pq .

For $\mathcal{A}$ to satisfy $\varphi(\bar{n}, \bar{c}, p\bar{c}, q, pq)$ there must be additional elements of $\mathcal{A}$ to witness the existential quantifiers in φ , denoted by $\bar{n}^* = \{m_1^*, \dots, m_{\ell^*}^*\} \subset N \setminus \bar{n}$ and by $\bar{c}^* = \{c_1^*, \dots, c_{k^*}^*\}$ in $Q \setminus (\bar{c} \cup p\bar{c} \cup \{q, pq\})$, such that $\mathcal{A}$ satisfies a conjunction of instances of $<$ and R or their negations about elements of $\bar{c}, p\bar{c}, \bar{n}, \bar{c}^*, \bar{n}^*, q$ and pq . Fix such $\bar{c}^*$ and $\bar{n}^*$ and let D be the set with elements $\bar{n}, \bar{c}, p\bar{c}, \bar{n}^*, \bar{c}^*, q$ and pq . Since there are only finitely many possibilities, without loss of generality, we may assume that the given conjunction includes all instances of the relations $<$ or R or of their negations which hold of elements of D . We call this the $\mathcal{A}$ -diagram of D .

By fixing the elements of D , we have also fixed all the ratios between the nonzero elements of $D \cap Q$, which determines not only which of their triples satisfy R but also by which multiplicative scalars. Specifically, when x is non-zero, $R(m, x, y)$ holds if and only if $y/x \in S_m$. Consider the $\bar{n}$ -graph on $D \cap Q$ in which two points x and y are connected by an edge if and only if there is a $m \in \bar{n}$ such that either both x and y are zero or one of x/y or y/x is an element of S_m . The $\bar{n}$ -component of an element of $x \in D \cap Q$ is the set of $y \in D \cap Q$ such that there is a path in the $\bar{n}$ -graph which goes from x to y . Let $C(\bar{n}, pq)$ denote the $\bar{n}$ -component of pq .

We observe that $C(\bar{n}, pq)$ does not include q or any element of $\bar{c} \cup p\bar{c}$. This is because multiplication of a rational number with reduced-form fractional representation a/b by an element of any S_m , $m \in \bar{n}$, can neither add nor delete a factor of p or q from a or b in the fractional representation of the product. Neither q nor any element of $\bar{c} \cup p\bar{c}$ is denoted by a fraction a/b in reduced such that both q and p divide a . But, by the same reasoning, all of the elements of $C(\bar{n}, pq)$ are denoted by such reduced-form fractions.

We exhibit a partially stretched copy $D^\dagger$ of D . Let r be a prime which is free over $q, p, \bar{c}$, all the $s_j \in S_m$, for $m \in \bar{n}$, and all other elements of D . First, replace pq by pqr and replace each element c_i^* of $C(\bar{n}, pq)$ with c_i^*r . In particular, do not replace q or any element of $\bar{c} \cup p\bar{c}$. Second, let $\bar{n}^\dagger$ be a sequence of indices which are pairwise ordered under $<$ in the same way as $\bar{n}^*$ is ordered and such that for each $m^* \in \bar{n}^*$, $S_{m^\dagger}$ is the smallest superset of S_{m^*} such that the following conditions hold.

1. For each x and y in D , if x is not in $C(\bar{n}, pq)$, y is in $C(\bar{n}, pq)$ and $R(m^*, x, y)$ then $ry/x \in S_{m^\dagger}$.

2. For each x and y in D , if x is in $C(\bar{n}, pq)$, y is not in $C(\bar{n}, pq)$ and $R(m^*, x, y)$ then $y/xr \in S_{m^\dagger}$.

Since each finite set S appears infinitely often as an S_j and all of the elements of $\bar{n}^*$ are larger than the maximum of $\bar{n}$, there is such a sequence. Replace each m^* in D by $m^\dagger$ in $D^\dagger$.

Thus, we have a map $\dagger : D \rightarrow D^\dagger$, which replaces the $\bar{n}$ -component of pq with a new component rescaled by multiplication by r and which augments the sets S_{m^*} to include the graphs of appropriately rescaled maps. Intuitively, we have stretched $C(\bar{n}, pq)$ by a factor of r and correspondingly stretched or contracted its connections to the rest of D , depending on their directions.

By inspection, $\dagger$ preserves membership in N and membership in Q .

By the definition of $\dagger$, for all m_1^* and m_2^* in $\bar{n}^*$, $m_1^* < m_2^*$ if and only if $m_1^\dagger < m_2^\dagger$. So, the relation $<$ is preserved by $\dagger$.

We examine all the cases to show the same for the relation R . Consider a triple (m, x, y) . Begin with the case $m \in \bar{n}$. Then, $m^\dagger$ is equal to m . There are three possible cases for pairs (x, y) in $D^\dagger$: (1) neither comes from $C(\bar{n}, pq)$, (2) one does and the other does not, or (3) both do. In Case (1), $x^\dagger = x$, $y^\dagger = y$ and so $R(m, x, y)$ if and only if $R(m^\dagger, x^\dagger, y^\dagger)$ is true on trivial grounds. In (2), $R(m, x, y)$ cannot hold, by the definition of $C(\bar{n}, pq)$. We observe that neither $R(m, x, ry)$ nor $R(m, rx, y)$ can be true statements about elements of $D^\dagger$. First, if $x \notin C(\bar{n}, pq)$, $y \in C(\bar{n}, pq)$ and $s \in S_m$, then r is free over sx but not free over ry , so $R(m, x, ry)$, equivalently $R(m, x^\dagger, y^\dagger)$, also cannot hold. The dual case, in which $x \in C(\bar{n}, pq)$ and $y \notin C(\bar{n}, pq)$, is similar. In (3), both x and y belong to $C(\bar{n}, pq)$. Then $x^\dagger = rx$ and $y^\dagger = ry$ and $R(m, x, y)$ is equivalent to $R(m, x^\dagger, y^\dagger)$, by the observation that R is invariant under scalar multiplication on the second and third coordinates.

Now, we examine the case $m^* \in \bar{n}^*$. By definition, $S_{m^\dagger}$ is equal to

$$S_{m^*} \cup \left\{ sr : \exists x, y \in D \begin{pmatrix} y \in \bar{n}\text{-component of } pq \text{ and} \\ x \notin \bar{n}\text{-component of } pq \text{ and} \\ s = y/x \in S_{m^*} \end{pmatrix} \right\} \\ \cup \left\{ s/r : \exists x, y \in D \begin{pmatrix} y \in \bar{n}\text{-component of } pq \text{ and} \\ x \notin \bar{n}\text{-component of } pq \text{ and} \\ s = x/y \in S_{m^*} \end{pmatrix} \right\}$$

Again, consider the three cases for x and y . For Case (1), suppose that neither x nor y belongs to $C(\bar{n}, pq)$. Then $x^\dagger = x$ and $y^\dagger = y$. For any m^* ,

since S_{m^*} is contained in $S_{m^\dagger}$, if $R(m^*, x, y)$ then $R(m^\dagger, x, y)$, which is the same as $R(m^\dagger, x^\dagger, y^\dagger)$. Since r is free over x and over y and the additional elements of $R(m^\dagger, x, y)$ involve multiplication by a rational number with a factor of r in either its denominator or in its numerator, if $R(m^\dagger, x^\dagger, y^\dagger)$, equivalently $R(m^\dagger, x, y)$, then $y/x \in S_{m^*}$ and $R(m^*, x, y)$.

For (2), suppose y is in $C(\bar{n}, pq)$ and x is not. By definition of $\dagger$, $x^\dagger = x$ and $y^\dagger = ry$. Suppose $R(m^*, x, y)$. Then $y/x \in S_{m^*}$, which implies $ry/x \in S_{m^\dagger}$, which implies $R(m^\dagger, x, ry)$, and so $R(m^\dagger, x^\dagger, y^\dagger)$. Conversely, suppose that $R(m^\dagger, x^\dagger, y^\dagger)$, equivalently $R(m^\dagger, x, ry)$. By the definition of R , $ry/x \in S_{m^\dagger}$. Since r is free over x , y and S_m , ry/x must have the form rs , for s in S_{m^*} , and so $y/x \in S_{m^*}$, equivalently $R(m^*, x, y)$. The case of $R(m^*, y, x)$ is similar, with division by r in place of multiplication by r .

Finally, suppose that both x and y belong to $C(\bar{n}, pq)$. Then $x^\dagger = rx$ and $y^\dagger = ry$ and neither is equal to zero. If $R(m^*, x, y)$, equivalently $y/x \in S_{m^*}$, then $ry/rx \in S_{m^*}$, equivalently $y^\dagger/x^\dagger \in S_{m^*}$. Since $S_{m^*} \subseteq S_{m^\dagger}$, $y^\dagger/x^\dagger \in S_{m^\dagger}$ and so $R(m^\dagger, x^\dagger, y^\dagger)$ holds. Conversely, suppose $R(m^\dagger, x^\dagger, y^\dagger)$. Then $y^\dagger/x^\dagger \in S_{m^\dagger}$, which is the same as saying $ry/rx \in S_{m^\dagger}$ and equivalent to $y/x \in S_{m^*}$. Since r is free over x and y , y/x must be in $S_{m^\dagger}$ by reason of $y/x \in S_{m^*}$. Thus, $R(m^*, x, y)$ must hold. This exhausts all the cases and proves that $R(m^*, x, y)$ holds if and only if $R(m^\dagger, x^\dagger, y^\dagger)$ also holds.

Thus, $\dagger$ is an isomorphism between the relational structures on D and $D^\dagger$. Quantifier-free formulas are preserved by such maps, so $\mathcal{A} \models \varphi(\bar{n}, \bar{c}, p\bar{c}, q, pqr)$, as was required to finish the proof of the theorem. $\square$

References

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